

The fourth generalized Davenport constant of C_5^3

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Abstract

For a finite abelian group G , the generalized Davenport constant $D_k(G)$ is the least ℓ such that every sequence over G of length at least ℓ contains k pairwise disjoint nonempty zero-sum subsequences. We determine the fourth generalized Davenport constant of the elementary abelian group C_5^3 :

$$D_4(C_5^3) = 30,$$

and consequently $D_k(C_5^3) = 5k + 10$ for every $k \geq 2$. The lower bound $D_k(C_5^3) \geq 5k + 10$ is a direct instance of Theorem 4.1 of Freeze and Schmid (Discrete Math. 310 (2010) 3373–3389), so the contribution of this paper is the matching upper bound: the Freeze–Schmid bound is tight for C_5^3 from $k = 2$ onward. The proof is finite and computer-assisted. The remaining case reduces to the nonexistence of a length-31 zero-sum sequence with no nonempty zero-sum subsequence of length at most five; a saturation argument restricts positive multiplicities to 1, 2, 4, the admissible support patterns form a 60-pattern family, and a rank/plane decomposition covers them by 78 exhaustive branches, each exhausted by two independently implemented exact searches with zero survivors in all 156 recorded runs. The branch cover was additionally regenerated by an independently written reconstruction that imports none of the original search code, and the acceptance pipeline rejects a battery of hostile mutations of the equations, the branch structure, and the result bindings; the exhaustive per-branch search itself was executed by the two original engines and a cross-machine replay of the same source, and has not been independently re-implemented. External human proof review, externally checked per-branch certificates, and final custody sign-off are deferred to journal review or later independent validation and are not represented as already performed.

1. Problem and theorem

Throughout, a sequence over C_5^3 is a finite multiset of group elements; it is *zero-sum* (or *total-zero*) if its elements sum to zero, and *short-zero-free* if it contains no nonempty zero-sum subsequence of length at most five, the exponent of C_5^3 . For $k \geq 1$, $D_k(G)$ is the least ℓ such that every sequence of length at least ℓ contains k pairwise disjoint nonempty zero-sum subsequences, and, following Fan, Gao, Wang, Zhong, and Zhuang (Electron. J. Combin. 19(3), P31, 2012), $C_0(G)$ denotes the set of lengths ℓ for which every total-zero sequence of length ℓ contains a nonempty zero-sum subsequence of length at most $\exp(G)$.

The lower-bound half of the picture is published prior work. Theorem 4.1 of Freeze and Schmid (Discrete Math. 310 (2010) 3373–3389) states that for $G = C_{n_1} \oplus \cdots \oplus C_{n_r}$ with $1 < n_1 \mid \cdots \mid n_r$,

integers $s \geq 2$ and $t \in [1, r]$ with $s(s-1)/2 \leq r-t+1$, and $\delta = 1$ if n_t is odd (else $\delta = 0$), one has $D_k(G) \geq D^*(G) + s\lfloor n_t/2 \rfloor + \delta + (k-2)n_r$ for $k \geq 2$, where $D^*(G) = \sum_i (n_i - 1) + 1$. Applied to $G = C_5^3$ with $s = 3$, $t = 1$ (the hypothesis holds with equality) and $\delta = 1$ since $n_t = 5$ is odd, this gives

$$D_k(C_5^3) \geq D^*(C_5^3) + 3\lfloor 5/2 \rfloor + 1 + (k-2) \cdot 5 = 5k + 10 \quad (k \geq 2).$$

Earlier work in this programme established the width-one upper companion

$$D_k(C_5^3) \leq 5k + 11 \quad (k \geq 4),$$

together with the implication, via the recurrence of Freeze and Schmid (Proposition 3.1.3 of the same paper) in an archived instantiation,

$$D_4(C_5^3) = 30 \implies D_k(C_5^3) = 5k + 10 \quad (k \geq 2).$$

Three further ingredients enter that implication, and we state their provenance explicitly so the full dependency chain is visible. $D_2(C_5^3) = 20$ follows from the published lower bound together with Zhao's short-subsequence lemma (arXiv:2506.21383, Lemma 4.4). $D_3(C_5^3) = 25$ and $s_{\leq 6}(C_5^3) = 24$ are established by exhaustive two-sided machine computations recorded in the research archive — for D_3 , an explicit length-24 witness with two but not three disjoint zero-sum subsequences, and an exhaustive pass finding zero length-25 sequences lacking three disjoint zero-sum subsequences — at the same computer-verified standard as the main computation of this paper, not as analytically proved theorems. The corridor itself is obtained from the base bound $D_4 \leq 31$ by the Freeze–Schmid recurrence at $\ell = 5$. Every ingredient of Theorem 1 below is therefore either cited published mathematics or machine-verified computation named here; nothing rests on an undisclosed assumption.

The one unresolved numerical question was therefore whether a length-31 total-zero sequence can be short-zero-free — equivalently, whether the Freeze–Schmid lower bound is attained.

Theorem 1. No length-31 total-zero sequence over C_5^3 is free of nonempty zero-sum subsequences of lengths one through five. Consequently $31 \in C_0(C_5^3)$, $D_4(C_5^3) = 30$, and $D_k(C_5^3) = 5k + 10$ for every $k \geq 2$.

The proof is finite and computer-assisted. The generic recurrence, localization, saturation, and projective-space results used as premises are prior mathematics and are not claimed here.

2. Finite grammar and branch cover

Let S be a hypothetical obstruction: a length-31 total-zero short-zero-free sequence. Since the support-size exclusion below gives $|\text{supp}(S)| \geq 14$, the saturation lemma of the companion corridor analysis applies (it requires support size at least nine) and restricts every positive multiplicity in S to 1, 2, 4.

Lemma (saturation defect; multiplicity 3 excluded). In a short-zero-free total-zero sequence of length 31 over C_5^3 with support size at least nine, no element occurs with multiplicity exactly $3 = p - 2$. *Proof sketch.* Five equal elements sum to zero, so every multiplicity is at most four. Append one further copy of an element x of multiplicity m : saturation produces a short zero-sum subsequence using the appended occurrence, and it must consume every original copy of x — otherwise an unused original copy yields a short zero-sum already inside S . For $m = 3$ the residual defect subsequence has length at most one and sum x , forcing another copy of x , a contradiction. The full proof is

Theorem 4 and Corollary 5 of the companion corridor manuscript in the research archive. If a_1, b_2, c_4 count the support points carrying those multiplicities, then

$$a_1 + b_2 + c_4 = s, \quad a_1 + 2b_2 + 4c_4 = 31.$$

Solving them gives $b_2 = 31 - s - 3c_4$ and $a_1 = 2s - 31 + 2c_4$; over the support range $14 \leq s \leq 31$ established below, nonnegativity alone yields exactly 60 admissible multiplicity patterns (42 with $s \leq 22$ and 18 with $s \geq 23$). Prior exact work excludes support size at most 13. The remaining support sizes 14–31 are covered by a rank/plane decomposition with 51 lower-support branches and 27 upper-support branches.

The exclusion of support size at most 13 is itself part of the archived programme: supports up to ten are exhausted by a bounded two-engine search, and supports 11–13 by an analytic case analysis over the nine admissible patterns there, recorded with its own theorem-level write-up in the research archive.

For support sizes 14–22 the cover argument is a three-case analysis on c_4 , the number of multiplicity-four points. If $c_4 \leq 2$, the high-multiplicity part carries mass at least $14 > 13 = \eta(C_5^2)$ (the known value, from $\eta(C_n^2) = 3n - 2$; see the survey of Gao and Geroldinger), forcing rank three. If $c_4 \geq 4$, four multiplicity-four points supply a rank-three basis. If $c_4 = 3$, the three multiplicity-four points contribute exactly 12 terms — one below the η threshold — so their rank is a genuine split: rank three gives a basis branch, while in the rank-two case they span a plane in which any multiplicity-two point would raise the plane mass to at least 14 and contradict η , forcing every doubleton (and, failing that, a singleton) outside the plane, since a length-31 total-zero sequence cannot have rank two. This yields 51 lower-support branches. For support sizes 23–31 the high-support set is partitioned by rank and multiplicity profile into four families (rank-three profiles keyed by c_4 ; a rank-two family admissible only at high-part mass at most 12; and the two low-height families), yielding 27 branches. The resulting cover contains exactly 78 branches; the complete case analysis, with every count, is written out in the independent branch-audit record of the research archive, and hostile perturbations of the analysis (omitting a branch, changing the total length, dropping the $c_4 = 3$ split, altering the η threshold) all change the branch count away from 78.

3. Exact search

The archived computation attacks each branch with two exact search engines built on different state representations: the primary engine keeps one 128-bit reachability mask per exact weight zero through five with coordinate-mask translations, while the second packs five 25-bit coordinate planes into vector lanes with separately written cyclic shifts. Three branch-family drivers times two representations give six sources. Each engine tracks all group sums reachable at exact weights zero through five and rejects a partial sequence as soon as zero becomes reachable in a forbidden weight layer. The two implementations share the branch specifications but not code; authorship is recorded in the archive by role rather than by name. Across the 78 branches and two engines, all 156 recorded runs complete, agree on the fixed branch fingerprints, and report zero solutions in every branch.

A replay on a second machine, using the same archived source but a different compiler major version and native instruction target, reproduces the exact result and cover digests. This replay is an independent execution of the same source; it is not counted as an independent implementation.

4. Independent reconstruction and adversarial verification

An independent reconstruction, written separately from the original search code, imports neither the original branch list, nor the C engines, nor their fingerprints or generator output. Starting only from the multiplicity equations and the rank/plane lemmas, it regenerates:

- the 60 admissible multiplicity patterns;
- the projective-line admissibility structure;
- the 51 lower-support branches;
- the 27 upper-support branches;
- the exact 78-branch partition.

Only after this reconstruction is complete is the regenerated branch set compared with the archived cover. The comparison is exact: no branch is missing and none is extra.

The reconstruction uses a materially different dynamic-programming representation in Python rather than a line-by-line translation of the original C search. In addition, an SMT-solver calibration exercises the primitive disjoint-zero-sum predicate on published small-group control values and agrees with a separately implemented brute-force occurrence assignment on exhaustive small domains. Where several verification routes exist, acceptance requires unanimity: a disagreeing or missing route terminates the check adversely or as unverifiable, never by majority vote.

Hostile controls confirm that acceptance depends on mathematical content rather than file identity: an omitted branch, an altered total-length equation, a deleted or modified rank split, an altered η threshold, a forged zero-solution digest, and a checksum-consistent forged nonzero-survivor result are all rejected.

5. Proof of Theorem 1

Assume a length-31 total-zero short-zero-free sequence exists. Saturation forces its positive multiplicities into $\{1, 2, 4\}$, hence its support pattern belongs to the 60-pattern family generated by the equations of Section 2. Prior work excludes support size at most 13. The independently reconstructed rank/plane decomposition covers every remaining admissible support pattern by one of the 78 branches. The archived exact computation exhausts every branch under two representations and records zero survivors in all 156 runs. Therefore the assumed obstruction does not exist, and $31 \in C_0(C_5^3)$. Combined with the previously established corridor, this gives $D_4(C_5^3) = 30$; the Freeze–Schmid recurrence (Proposition 3.1.3), in the archived instantiation, then yields $D_k(C_5^3) = 5k + 10$ for every $k \geq 2$. $\square$

6. Priority and novelty scope

A dedicated priority audit, refreshed on 2026-09-02 with programme-internal vocabulary stripped from every query, found no retrieved source that already establishes the exact value $D_4(C_5^3) = 30$ or any exact $D_k(C_5^3)$, and found that the lower bound $D_k(C_5^3) \geq 5k + 10$ ($k \geq 2$) is a direct instance of Freeze–Schmid Theorem 4.1, as stated in Section 1; the dated audit record accompanies the research archive. Two facts support the exact value being otherwise open: $\eta(C_5^3)$ has no published exact value (Fan–Gao–Zhong, J. Number Theory 131 (2011), prove $s(C_5^3) = \eta(C_5^3) + 4$ without determining either constant), so no exact $D_k(C_5^3)$ follows from the known upper-bound machinery; and the current state of the art (Zhong, Combinatorica 2025) records exact D_k values only for rank at most two, elementary 2-groups of rank at most five, and C_3^3 . MathSciNet and zbMATH were not accessible to this audit and remain explicitly unchecked; the audit closes the local literature task

but is not an omniscient global priority certificate. The paper therefore states the exact theorem and its relation to the closest retrieved work while avoiding absolute “first”, “unique”, or equivalent language.

For context, exact D_k values are currently published for groups of rank at most two (a linear formula), elementary 2-groups of rank at most five (Freeze–Schmid; ranks up to three earlier), and C_3^3 ; the present value is, to our retrieval, the first exact D_k determination for an odd- p elementary group of rank three beyond $p = 3$. Beyond the single value, the 60-pattern/78-branch reduction is reusable as a template for exact generalized-Davenport computations in other rank-three groups. The nearest published exact rank-three companion sharpens what is new here. For C_3^3 , the known values are $D_1 = 7$ (Olson’s $D = D^*$ for p -groups), and $D_2 = 11$ with $D_k = 3k + 6$ for $k \geq 3$ (Bhowmik and Schlage-Puchta 2007, Proposition 1, as recorded in Freeze–Schmid Remark 5.3): the Freeze–Schmid bound $D_k(C_3^3) \geq 3k + 5$ is attained at $k = 2$ but strictly exceeded from $k = 3$ onward. The present theorem shows the opposite regime for C_5^3 : the Freeze–Schmid bound is attained for every $k \geq 2$. Where the tightness threshold sits as a function of p for C_p^3 is, to our retrieval, open; the $p = 3$ versus $p = 5$ contrast is stated as data rather than explained.

The paper does not claim novelty for the generic Davenport machinery, the saturation argument, the projective-space methods, the recurrence, or exact-search methodology in the abstract, and it does not claim the lower bound, which is Freeze–Schmid’s. The contribution claimed here is the matching upper bound and hence exactness — the exact bounded theorem, its finite branch reduction, and the independently reconstructed verification path for this instance.

7. External-review boundary

The following items remain intentionally external and are **not** represented as locally completed:

1. a signed independent mathematical proof review by a person outside the authoring process;
2. externally checked per-branch certificate and output manifests;
3. final independent custody/authorization receipts where a venue or archive requires them;
4. journal peer review and editorial acceptance.

These items are deferred to journal review or later independent validation. Their absence does not erase the locally established theorem or block completion of the local submission package, but it does block any claim that external peer review, external custody, or independent human proof validation has already occurred.

8. Reproducibility and evidence map

The released research archive contains the complete evidence for every claim above: the exact-computation authority packet (`evidence/global-obstruction-v1/`), the independent reconstruction material (`evidence/a1-independent-branch-audit-v1/`), the dated priority-audit record (`PRIORITY_AUDIT_ADDENDUM_2026-09-02.md`), and the archive-readiness machinery (`evidence/a1-theorem-packet-archive-v1/`). The archive machinery is designed to fail closed when external review or custody receipts are absent; that fail-closed status is preserved rather than bypassed.

The local submission posture is therefore:

- exact theorem: established by the archived finite computation and proof chain;
- independent implementation/reconstruction: locally complete;

- mutation/adversarial controls: locally complete;
- priority audit: locally complete, with absolute priority language withheld;
- external human proof review/custody: deferred and explicitly unclaimed;
- editorial acceptance: unclaimed.

9. Conclusion

The remaining one-unit branch for the fourth generalized Davenport constant of C_5^3 is closed: $D_4(C_5^3) = 30$, and the Freeze–Schmid recurrence extends this to $D_k(C_5^3) = 5k + 10$ for every $k \geq 2$ — the Freeze–Schmid lower bound is tight for C_5^3 from $k = 2$ onward. The result rests on an exhaustive 60-pattern/78-branch reduction, two exact search representations, an independently regenerated cover, solver calibration, cross-machine replay, and hostile semantic controls. External human proof audit, external certificate checking, and journal review remain visible deferred validation rather than artificial blockers on local completion.

Related work and novelty boundary

The generalized Davenport constants ask for the least sequence length forcing a prescribed number of pairwise disjoint nontrivial zero-sum subsequences. For rank-two groups an exact linear formula is known, and recent inverse work studies the extremal structure at that solved boundary [Zhong 2025]. For general finite abelian groups the sequence is eventually linear in the index [Freeze–Schmid 2010], but the rank-three elementary setting is not covered by the rank-two formula. Nearby rank-three work studies other zero-sum invariants rather than the same generalized Davenport constant [Zhang 2023]. Standard background is provided by the zero-sum survey of Gao and Geroldinger and the monograph of Geroldinger and Halter-Koch.

The lower-bound half of the paper’s formula is published prior work: Theorem 4.1 of Freeze and Schmid, instantiated at $s=3, t=1, \delta=1$, gives $D_k(C_5^3) \geq 5k+10$ for every $k \geq 2$. The exact rank-three companion C_3^3 [Bhowmik–Schlage-Puchta 2007, recorded in Freeze–Schmid Remark 5.3] attains that bound only at $k=2$ and exceeds it from $k=3$ onward. No exact value of $\eta(C_5^3)$ is published [Fan–Gao–Zhong 2011 prove $s(C_5^3)=\eta(C_5^3)+4$ without determining either constant], so no exact $D_k(C_5^3)$ follows from the known upper-bound machinery.

The current paper’s contribution is the matching upper bound and hence exactness: it proves $D_4(C_5^3)=30$, so the Freeze–Schmid lower bound is tight for C_5^3 from $k=2$ onward — the opposite tightness regime from C_3^3 , stated as data. The residual novelty claim is confined to this bounded computer-assisted theorem and its checked decomposition; it is not a new general recurrence theory, and the lower bound is not claimed.

The primary-source review was refreshed on 2026-09-02 with a PARTIAL verdict (record: `PRIORITY_AUDIT_ADDENDUM_2026-09-02.md`): no retrieved source states the exact value $D_4(C_5^3)=30$ or any exact $D_k(C_5^3)$; MathSciNet and zbMATH were not accessible and remain CANNOT_CHECK. Accordingly the paper makes no “first”, “unique”, or exhaustive-nearest-work claim; a closer source found during editorial review should narrow positioning without changing the checked theorem statement.

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